

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Design a milling fixture for cutting an internal 4 splines of size 100x10x5 mm inside a hollow shaft of 30 mm internal diameter and 60 mm external diameter.
- Q.24 Design and draw a snap gauge for measuring a flat to flat distance of a hexagonal nut of bolt diameter of 20 mm.
- Q.25 Explain any four types of drilling jigs with neat sketches.

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4th Sem / Mechanical (Tool & Die Design)

Subject : Jigs, Fixtures and Gauges - Design and Drawing

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Total cost of production_____ with the use of jigs and fixtures
- a) Increases
 - b) Decreases
 - c) Remains same
 - d) Jigs are never used for production
- Q.2 Which machining operation is a jig typically NOT used for
- a) Drilling
 - b) Reaming
 - c) Tapping
 - d) Milling
- Q.3 A fixture does not
- a) Holds the workpiece
 - b) Locate the workpiece
 - c) Guide the tool
 - d) None of the above

- Q.4 Principle of _____ states that “in order to achieve the maximum accuracy in locating point should be placed as far apart from one another as it is possible”
- Six point location
 - Least point
 - Extreme position
 - Mutually perpendicular planes
- Q.5 Which type of jig is used for drilling a series of equi distance holes on a circle?
- Index jig
 - Plate jig
 - Box jig
 - Universal jig
- Q.6 Why the diamond shape of the locating pin advantageous in jigs and fixtures?
- It permits major adjustments in the workpiece position
 - It provide increases surface area for clamping.
 - It allows for easier assembly and disassembly of the jig
 - It reduces wear and tear due to its self aligning feature

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Plug gauge is a limit gauge.(True/False)
- Q.8 Indexing is used for _____.
- Q.9 Bushes need to _____. (Hard/ Soft)

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- Q.10 “NOT GO” Size of a hole gauge has diameter equal to _____
- Q.11 The fixture are _____ than the jigs.(lighter/heavier)
- Q.12 The material used for gauges is _____

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe the factors to be considered for material selection for jigs and fixtures.
- Q.14 Draw free hand a gang milling fixture.
- Q.15 Write a short note on selection of gauges for inspection.
- Q.16 Classify various type of fixtures.
- Q.17 Differentiate between latch type and channel type fixtures.
- Q.18 Describe any four locating methods.
- Q.19 Draw any one lathe fixture.
- Q.20 Write short note on mass manufacturing with example.
- Q.21 Explain basic principles of locating jobs.
- Q.22 Describe how maximum and minimum material condition is used in gauges?

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